

DEREK CHIBUZOR

Los Angeles, CA | (201) 316-3229 | derekchibuzor777@gmail.com | [linkedin.com/in/derekchibuzor](https://www.linkedin.com/in/derekchibuzor) | derekc22.github.io

EDUCATION

University of Southern California	Los Angeles, CA
Master of Science in Mechanical Engineering	Aug. 2025-Dec. 2026
* GPA: 3.90/4.00 Robot Dynamics & Control, Linear Systems, Flight Vehicle Stability & Control, Optimization Methods	
Bachelor of Science in Aerospace Engineering	Aug. 2021-May 2025
* GPA: 3.62/4.00 Computer-aided Design, Dynamic Systems, Linear Control, Computational Methods, Flight Mechanics	

WORK EXPERIENCE

Neros Technologies	Torrance, CA
Controls Engineer, Autonomy	Jan. 2026-Present
* Develop performant guidance, navigation, and control algorithms for deployment on compute-constrained UAV platforms.	
* Contribute to internal 6-DoF flight simulator with custom dynamics, control, state estimation, and perception modules.	
* Conduct hardware-in-the-loop (HIL) testing to tune controllers, perform system identification, and characterize software performance.	
Dynamic Robotics and Control Laboratory	Los Angeles, CA
Research Assistant, Sim2Real	Jan. 2025-Present
* Engineered system identification framework for Sim2Real transfer via gradient- and sampling-based hybrid optimization .	
* Trained evolutionary algorithms and machine learning models to improve actuator dynamics modelling by 78% in 24-DoF robot.	
* Constructed diffusion model framework for generating full-body loco-manipulation trajectories from Unitree G1 motion capture data.	
Lawrence Livermore National Laboratory	Livermore, CA
Computational Engineer Intern	June 2025-Aug. 2025
* Wrote IK task-space PD controller for 6-DoF UR3e, saving 5+ hours of manual path planning while achieving 0.1 mm tracking error.	
* Trained reinforcement learning policies for pick and place tasks, reducing episode length by 20% with transformer feature extractor.	
* Utilized implicit and explicit structural dynamics codes to perform modal analysis of jointed beam subject to broadband excitation.	
Northrop Grumman	Roy, UT
Mechanical Engineer Intern	June 2023-Aug. 2024
* Constructed 2-DoF reduced-order Simulink model of subterranean shock-isolated system , saving 240 hours of FEA computation.	
* Designed CAD assemblies, performed FEA , and wrote RFIs for elastomeric shock isolators and maintenance access doors.	
Amazon & Information Sciences Institute	Los Angeles, CA
GNC Engineer Intern	June 2022-Aug. 2022
* Developed vision-based navigation software, enabling pose estimation for 3-DoF rendezvous and proximity operations (RPO) .	
* Achieved sub-160 ms latency localization and tracking of 12 infrared LED targets up to 2.50 m from Raspberry Pi NoIR camera.	

SKILLS & CREDENTIALS

- * **Programming:** Python, C++, MATLAB, Java, JavaScript
- * **Software:** Simulink, Siemens NX, PyTorch, MuJoCo, CasADi, JAX, OpenCV, LabVIEW, Betaflight
- * **Hardware:** Raspberry Pi, Arduino

PROJECTS

- * **6-DoF Flight Simulator:** C++ flight simulator. Custom dynamics, control, state estimation, etc. with FlightGear integration.
- * **Multi-Agent DMPC:** Distributed and decentralized model predictive control (MPC) for high-DoF multi-agent systems.
- * **Dual-Axis Control System:** Sun-seeking, multithreaded, 2-DoF electromechanical solar array articulation module for 3U CubeSat.